ABSTRACT

BACKGROUND:

Stenosing tenosynovitis or trigger finger is characterised by pain, swelling, movement limitation, and a triggering sensation of the finger. Percutaneous release of the A1 pulley results in a low complication rate and high patient satisfaction.

OBJECTIVE:

- To compare the frequency of cure and relapse of percutaneous A1 pulley release with conventional open surgery in treatment of the trigger finger

STUDY DESIGN:

- Randomized Controlled Trial

SETTINGS:

- Orthopaedics Department, Benazir Bhutto Hospital, Rawalpindi

DURATION OF STUDY:

- 6 months after the approval of synopsis

  From: 30th April 2020 to 30 Sept 2020

METHODOLOGY:

After taking written informed consent, patients meeting the inclusion criteria were randomly assigned to either group; Group A (Percutaneous release) and Group B (Open surgery) by lottery method.

Group A was treated with a percutaneous release of the affected first annular pulley under local anesthesia and without any corticosteroid injection. In addition, group B underwent a conventional open surgical release of the A1 pulley, through a 10-15 mm incision. Demographic information (including name, age and gender) was also recorded.

RESULTS:

In our study, the frequency of cure and relapse of percutaneous a1 pulley release with conventional open surgery in treatment of the trigger finger was recorded as 86.67%(n=26) in Group-A and 96.67%(n=29) in Group-B cure while 13.33%(n=4) in Group-A and 3.33%(n=1) in Group-B were relapse, p value was 0.16.

CONCLUSION:

- We found no significant difference in frequency of cure and relapse of percutaneous A1 pulley release with conventional open surgery in treatment of the trigger finger.

KEYWORDS:

- *Trigger finger, percutaneous A1 pulley release, open surgery, cure and relapse*

INTRODUCTION

Stenosing tenosynovitis with mechanical impingement of the flexor tendons at the A1 pulley is a common condition affecting the digits in the following order of decreasing prevalence: Thumb, ring, middle, small and index. A nodule or thickening in the flexor tendon becomes trapped proximal to the pulley, making finger extension difficult. There have been described several management approaches in the treatment of trigger finger disease.

Nonsurgical management includes corticosteroid injection and splinting. However, according to a Fleisch et al[1] corticosteroid injections are effective in just 57% of patients. Additionally, that technique can result in upto 29% recurrence reported by Nikolaou et al.[2] Furthermore, many patients treated by conservative means, like immobilization or injections, may additionally require surgery.[3] In a recent clinical trial, designed to compare the effectiveness of 2 splint designs in treating trigger finger by Tarbhai et al reported,[4] the results showed positive outcomes in 50%-77% of the patients, depending on the type of splinting. On the other hand, Wang et al supported the superiority of surgical procedures for the definite treatment of the disease.[5]

Indeed, the percutaneous and open surgery methods have been proved more efficient than simple corticosteroid injection, regarding the cure and relapse of the disease.[5-6] Conventional open surgical technique remains the gold standard of treatment options.[7] However, surgical treatment also has complications, including scarring, surgical site infections and nerve injuries, in addition to possible disease relapse.[8]

Percutaneous surgical release of trigger finger is a preferable alternative to open surgery,[9] although a potential disadvantage can be the injury to either nerve or tendon due to the limited visibility.[7] Many hand surgeons avoid this treatment option due to the close proximity to the visibility.[7] Many hand surgeons avoid this treatment option due to the close proximity to the digital nerve.[2]

According to a meta-analysis of current literature by Zhao et al[11] blind percutaneous release has become more and more population lately, with overall increased success rates. Nikolaou et al[2] reported 93.75% cure and 6.25% relapse with USG percutaneous release and 100% cure and 0% relapse with open surgery, cosmetic results found excellent or good in 87.5% in percutaneous release patients, while only 56.25% patients were evaluated as faire or poor in open surgery.

There is lack of randomized clinical trials in Pakistan, comparing open surgery to percutaneous release. In the present randomized controlled trial, we will try to compare the frequency of cure of percutaneous A1 pulley release with conventional open surgery in terms of ability to correct the trigger finger so that better management be planned.

REVIEW OF LITERATURE

DEFINITION

Trigger finger is the snapping, triggering, or locking of the finger as it is flexed and extended. Thickening and disproportionate narrowing of the retinacular sheath relative to its flexor tendons occur due to hypertrophy and fibrocartilaginous metaplasia at the tendon-pulley interface.[11] While known as stenosing tenosynovitis, tenovaginitis is a more accurate term as the histopathologic changes localize to the retinacular sheath and peritendinous tissue rather than the tenosynovium.

Normal tendon glide is most prominently affected at the A1 pulley where there is greater angulation of the tendon as it enters the pulley system. Trigger finger is thought to arise from high pressures at the proximal edge of the A1 pulley, the most common site of triggering, at the level of the metacarpal head.[12]

The thumb (33%) and the ring finger (27%) are most commonly affected in adults, but 90% of pediatric trigger fingers involve the thumbs, 25% of which are bilateral.[12] Pediatric trigger thumb occurs more frequently as a result of focal enlargement of the flexor pollicis longus, but no definite ultrasound abnormality of the A1 pulley has been noted.[13] Primary idiopathic trigger finger is more common while secondary trigger finger is associated with diabetes mellitus, rheumatoid arthritis, hypothyroidism, histiocytosis, amyloidosis, and gout.[12]

Incidence is generally thought to be 2% in the general population, more common in women and in patients with diabetes (7%) and rheumatoid arthritis.[14] The relationship of trigger finger to repetitive trauma has been cited in the literature; however, the exact mechanism of this correlation is still open to debate. Rarely, it is due to acute trauma or space-occupying lesions.[15]

EPIDEMIOLOGY:

Trigger finger is one of the most common causes of hand pain in adults. The reported prevalence is roughly 2 percent in the general population, and is most common among women in the fifth or sixth decade of life.[16] It can occur in one or more fingers in each hand and can be bilateral. The prevalence of trigger finger is also higher among patients with diabetes mellitus, rheumatoid arthritis, or conditions that cause systemic deposition of protein such as amyloidosis.[16-17]

Trigger finger is occasionally observed in children; in the pediatric age group, there is a much higher probability of an anatomic variation or an inherited condition.[17]

ASSOCIATED DISEASES:

Overuse, diabetes, gout, acromegaly, renal disease, glycogen storage diseases, carpal tunnel syndrome, rheumatoid arthritis, and other rheumatoid and musculoskeletal disorders have been associated with TF,.[18] Thyroid dysfunction, particularly hypothyroidism and thyrotoxicosis, have also been associated with TF.5 Lifetime risk of TF for the general population is 2.6% compared with 10% for people with diabetes.15 Others report rates of 1%–2% for the general population, and 10%–20% for people with diabetess; 25% of patients presenting with TF are diabetic.[19]

Furthermore, around half of patients with diabetes with TF will present with multiple digit involvement.[20] The longer a patient has diabetes, and specifically a high HbA1c, the more likely they will be affected by a hand or shoulder disorder. An HbA1c level greater than 7% is an independent risk factor for the development of TF, but an HbA1c level beyond 7% does not further increase risk. One report showed 60% of TF cases in patients with diabetes to recover spontaneously, compared with only 20%–29% of all TF cases.[20]

Carpal tunnel release (CTR) has been associated with the development of TF; the most common finger that triggers following a CTR is the thumb. CTR is accomplished by the release of the flexor retinaculum; however, this leads to the bowstring effect. The bowstring effect theoretically causes an increased friction force on the flexor tendons, putting patients at risk for TF. Within the first 6 months, there is a 9.65-fold increase in risk, and overall a 3.63-fold higher risk, when compared with the general population.[18]

In addition, patients with diabetes are more likely to develop TF following a CTR compared with patients without diabetes. For patients with diabetes, 8% will develop TF within 6 months of a CTR, and 10% within a year, compared with 3% and 4%, respectively, for non-diabetics. CTR involving removal of the forearm fascia has a greater risk for postoperative TF compared with transverse carpal ligament release alone due to an increased entrance angle of the flexor tendons into the A1 pulley with resulting friction and ultimately triggering events.[21]

Research that supports an increased risk of TF following CTR uses the contralateral hand as a control. Carpal tunnel syndrome and TF have common pathologies and are often diagnosed concomitantly in the same hand. CTR may not increase the risk of developing TF in the operative hand, but rather there is an intrinsic risk of developing TF in hands with carpal tunnel syndrome.[22]

SIGNS AND SYMPTOMS OF TRIGGER FINGER

Signs and symptoms of TF are as follows:

- Locking or catching during active flexion-extension activity (passive manipulation may be needed to extend the digit in the later stages)
- Stiff digit, especially in long-standing or neglected cases
- Pain over the distal palm
- Pain radiating along the digit

- Triggering on active or passive extension by the patient
- Palpable snapping sensation or crepitus over the A1 pulley
- Tenderness over the A1 pulley
- Palpable nodule in the line of the flexor digitorum superficialis (FDS), just distal to the metacarpophalangeal (MCP) joint in the palm
- Fixed-flexion deformity in late presentations, especially in the proximal interphalangeal (PIP) joint
- Evidence of associated conditions (eg, rheumatoid arthritis [RA], gout)
- Early signs of triggering in other digits (may be bilateral)

Children with trigger thumb rarely complain of pain. They usually are brought in for evaluation when aged 1-4 years, when the parent first notices a flexed posture of the thumb's interphalangeal (IP) joint. These children often demonstrate bilateral fixed flexion contractures of the thumb by the time they present to the physician.[23] By the time the child presents to the clinic, surgical treatment is already indicated in most instances.

ANATOMY

Fingers

Tendon sheaths of the long flexors run from the level of the metacarpal heads (distal palmar crease, superficial; volar plate, deep) to the distal phalanges. They are attached to the underlying bones and volar plates, which prevent the tendons from bowstringing. Predictable and efficient thickenings in the fibrous flexor sheath act as pulleys, directing the sliding movements of the fingers.

The two types of pulleys are anular (A) and cruciate (C). Anular pulleys are composed of single fibrous bands (ie, rings), whereas cruciate pulleys have two crossing fibrous bands.

The order of the pulleys from proximal to distal is as follows:

- The A1 pulley overlies the MCP joints; it is released during surgery for TF (Fig. 1)
- The A2 pulley overlies the proximal end of the proximal phalanx
- The C1 pulley overlies the middle of the proximal phalanx
- The A3 pulley lies over the PIP joint
- The C2 pulley lies over the proximal end of the middle phalanx
- The A4 pulley lies over the middle of the middle phalanx

- The C3 pulley lies over the distal end of the middle phalanx
- The A5 pulley lies over the proximal end of the distal phalanx

The A2 and A4 pulleys are vital in preventing bowstringing of the flexor tendons and must be preserved or reconstructed after any damage to them.

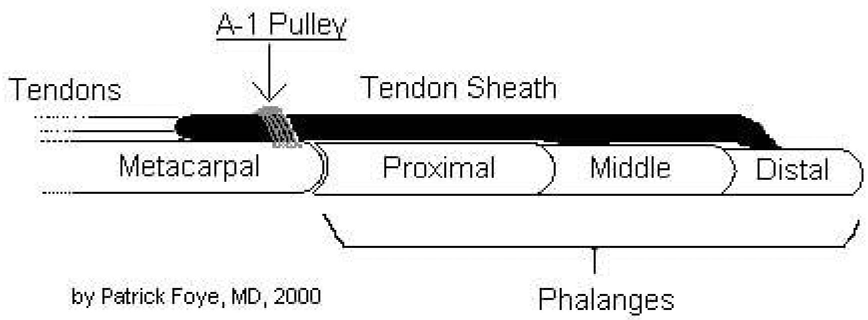

Fig. 1

Flexor tendons pass within tendon sheath and beneath A1 pulley at approximately metacarpal head, beyond which they travel into digit.

Thumb

The flexor anatomy of the thumb differs from that of the fingers. The flexor pollicis longus (FPL) tendon is a single tendon within the flexor sheath that inserts onto the base of the distal phalanx. The fibro-osseous sheath is composed of two anular pulleys (A1 and A2) that arise from the palmar plates of the MCP and IP joints, respectively. The oblique pulley, which originates from and inserts onto the proximal phalanx, is the most important pulley from a biomechanical perspective. The oblique pulley is approximately 10 mm in length, blending with a portion of the adductor pollicis insertion.

The digital nerves and arteries run parallel to the tendon sheath distally. At the level of the MCP flexion crease, they lie just deep to the skin. Proximal to the A1 pulley, the radial digital nerve of the thumb crosses obliquely over the sheath.

TISSUE ARCHITECTURE

Ultrasonography has demonstrated that untreated TF has on average an A1 pulley thickness of 1.1 mm; 1 month following corticosteroid treatment, the A1 pulley decreased to 0.7 mm for intrasheath injection and 0.8 mm for extrasheath injection.[24]

Injection site does not significantly affect the outcome. In addition to a thicker A1 pulley, a thicker proximal region of the A2 pulley and flexor tendon is also characteristic of TF, and within a month of corticosteroid treatment, tendon thickness significantly decreases. Tendon thickness decreased from 4.1 mm (for intrasheath) and 4.0 mm (for extrasheath) to 3.9 and 3.8 mm, respectively; following corticosteroid treatment, both flexor digitorum tendon and A1 thickness were comparable to control values.[25] It is speculated that corticosteroids have a more immediate effect on the A1 pulley than the tendon due to differences in tissue density. Tendon rupture may occur with repeated high-dose therapy.[26]

ETIOPATHOGENESIS

The majority of trigger fingers are idiopathic. Symptoms usually begin spontaneously, without a prior history of trauma or change in activity level. There are some observational reports suggesting an association with occupational or repetitive activities, but this is controversial.[27]

The main histopathological change is fibrocartilaginous metaplasia of the ligamentous layer of the tendon sheath at the first annular (A1) pulley with secondary reduction in the cross-sectional area of the fibro-osseous canal.[27] Functional impairment in finger flexion and extension is primarily a result of mechanical impingement leading to tendon entrapment.

PATHOPHYSIOLOGY:

A mismatch between the flexor tendon and the proximal pulley mechanism occurs in most cases of trigger finger. Normally, the tendons of the finger flexors glide back and forth under a restraining pulley.[28-30] Thickening of the flexor tendon sheath restricts the normal gliding mechanism. A nodule may develop on the tendon, causing the tendon to get stuck at the proximal edge of the A1 pulley when the patient is attempting to extend the digit, thereby causing difficulty. (Fig. 2)

When more forceful attempts are made to extend the digit, by using increased force from the finger extensors or by applying an external force (for example, by exerting force on the finger with the other hand), the digit classically snaps open with significant pain at the distal palm and into the proximal aspect of the affected digit. Less commonly, the nodule is restricted distal to the A1 pulley, resulting in difficulty flexing the digit.

Using sonoelastography, a newer technique for quantitative assessment of the stiffness of soft tissues, the data from one study noted that the causes for snapping in TF were increased stiffness and thickening of the A1 pulley. Three weeks after corticosteroid injection, the pulley thickness and the ratio of subcutaneous fat to the pulley both decreased; snapping disappeared in all patients studied.[31]

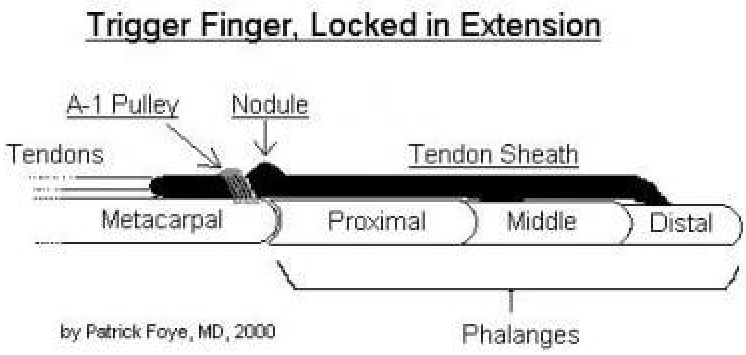

Fig. 2

Inflamed nodule can restrict tendon from passing smoothly beneath A1 pulley. If nodule is distal to A1 pulley (as shown in this sketch), then digit may get stuck in extended position. Conversely, if nodule is proximal to A1 pulley, then patient's digit is more likely to become stuck in flexed position.

PRESENTATION

Patients with trigger finger initially describe painless snapping, catching, or locking of one or more fingers during flexion of the affected digit. This often progresses to painful episodes in which the patient has difficulty spontaneously extending the affected digit(s). The pain is localized over the volar aspect of the metacarpophalangeal (MCP) joint and radiates into the palm or the distal finger.

The patient may rub over the tendon in the palm or demonstrate the locking phenomenon when describing the condition. Some patients awaken with the finger locked in the palm, with gradual "unlocking" as the day progresses.

In more severe cases, the finger may become locked in flexion requiring passive manipulation of the finger into extension, which can be painful. Reluctance to fully flex and extend the digit because of pain or locking can eventually lead to secondary contracture at the proximal interphalangeal (PIP) joint. It is also not unusual for a patient to have multiple trigger digits.

HISTOPATHOLOGICAL CHANGES

The A1 pulley comprised 3 layers. The outer layer is a highly vascularized convex layer that is continuous with the tendon sheath. The inner 2 layers are avascular and function as the concave gliding surface for the tendon; the first layer contains cartilage-like cells, and the second has spindle-shaped fibroblasts with elongated nuclei and compact parallel regular collagenous bundles.[32] The inner fibrocartilage becomes thinned and replaced by fibrous tissue in moderate TF.

Pathological transformation begins with a myxoid matrix between collagen and evolves to an irregular distribution of chondromyxoid matrix with vascular hyperplasia. Normal elongated nuclei of fibroblasts are replaced by rounded nuclei of chondrocytes. The accumulation of hyaluronic acid, chondroitin sulfate, and proteoglycan is associated with syndrome severity. Pathologically, the highest grade of TF contains invasive chondroid metaplasia. Computer grading using abnormal tissue and round nuclei as the pathological parameters strongly correlates with clinical severity and pathological grading.[32]

Evidence also suggests tendinosis of the trigger tendons; tissue samples of the tendons in TF had a Movin Score 14.2 compared with 2.5 for normal finger tendons. Trigger tendons also show significant upregulation of collagen types 1a1 and 3a1, aggrecan, and bigylcan, and downregulation of MMP-3 and TIMP-3.[33]

The histopathology of the tenosynovium surrounding the A1 pulley is also abnormal. It was found that 61% of samples comprised hyaluronic acid producing chondrocytoid cells that express CD44 (a marker for synovial B cells), but not S100 (normal chondrocyte cell surface antigen). Additionally, a hypocellular collagen matrix is observed in 84% of TF tenosynovium. This edematous tissue most likely contributes to the pressure between the A1 pulley and the tendon, contributing to the progression of TF. Inflammatory infiltrate, increased vascularity, hyperplasia of synovial lining, and fibrin exudation, which are markers of synovitis, are present in only 5%, 37%, 37%, 21%, and 5% of tissue.[34]

DIAGNOSIS:

The diagnosis of trigger finger is primarily based upon a history of locking or clicking during finger movement, which can be demonstrated on physical examination when the patient is asked to fully open and close the hand. The hands are placed in the palms-up position and the patient is asked to actively flex and extend the fingers, and to try to make the finger lock or catch (Fig 3). Alternatively, if active triggering is not present, the examiner places his fingers on the proximal interphalangeal joint (PIP) as the finger is actively flexed and extended, noting the presence of loss of smooth motion or a clicking sensation. The locking does not have to occur with every repetition.

Additional findings may include pain or tenderness at the base of the finger, directly over the tendon as it courses over the metacarpophalangeal (MCP) joint. A tender nodule (thickening of the tendons) may also be present. Pain may be aggravated by stretching the tendon in extension or by resisting flexion isometrically (Fig. 4). In addition to the nodule or swelling of the tendon, patients may also have a palmar nodule of fascial thickening overlying the tendon sheath associated with a concurrent Dupuytren's contracture.

Radiographs are not necessary in diagnosing patient with suspected trigger finger.[35]

Fig. 3

Mechanical triggering with flexor tenosynovitis (trigger finger)

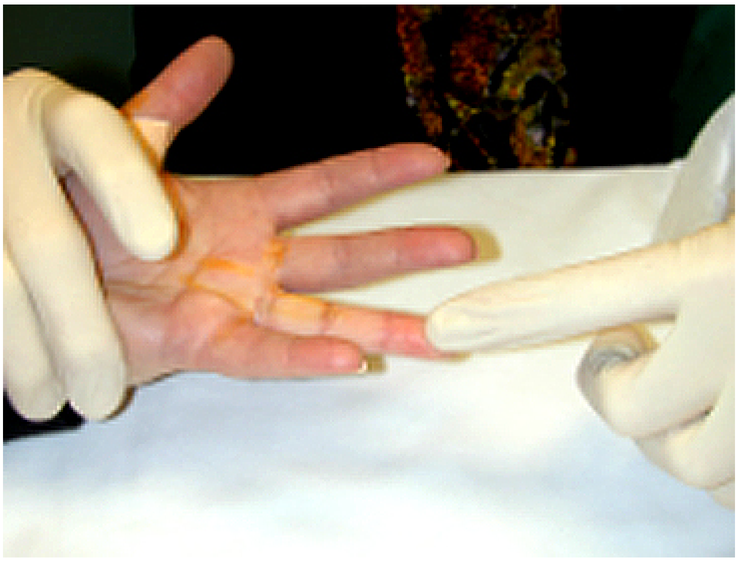

Fig. 4

Passive stretching sign for active tenosynovitis of the finger (trigger finger)

DIFFERENTIAL DIAGNOSIS

The differential diagnosis of trigger finger includes other conditions that can lead to locking, pain, loss of motion, and swelling of the metacarpophalangeal (MCP) joints, several of which are presented below.

Dupuytren's contracture:—

Dupuytren's contracture is characterized by loss of full extension of the affected finger or fingers at the MCP joint. The loss of extension is fixed and chronic, while in a trigger finger it is dynamic and episodic. Unlike trigger finger, Dupuytren's is painless and nodular lesions are typically evident in the palmar fascia. The nodule typically progresses over a period of time to form a fibrous cord that extends from the palm to the digits, which also helps distinguish it from trigger finger.

Diabetic cheiroarthropathy:—

Diabetic cheiroarthropathy is characterized by a typically painless, limited ability to fully flex or extend the MCP joints and/or interphalangeal joints. It generally affects all fingers symmetrically, whereas trigger finger may affect one or, less often, more fingers. It may progress to fixed flexion contractures of the finger joints. A tight, waxy skin surface over the dorsum of the hand is usually observed, which is not observed in trigger finger. However, patients who have diabetic cheiroarthropathy and limited joint mobility are more likely to also develop trigger finger.[36]

Metacarpophalangeal joint sprain:—

Tenderness on either side of the MCP joint associated with loss of full flexion suggests a sprain of the MCP joint, particularly in the setting of recent trauma. Thus, a history of trauma and the absence of triggering can help distinguish this from trigger finger.

Infection within the tendon sheath:—

Severe pain and tenderness of the flexor tendon in the hand, particularly in the setting of a preceding puncture or bite wound involving the finger or hand suggests infectious tenosynovitis. It is imperative to recognize this syndrome, since an unrecognized closed-space infection can lead to severe limitation of motion due to tendon disruption. It can be distinguished from trigger finger as it often leads to more severe pain, tenderness, erythema, and swelling in long axis over the affected tendon or finger. Infection of the hand requires immediate surgical consultation for possible exploration and drainage.

Calcific peritendinitis or periarthritis:—

Calcific peritendinitis or periarthritis is an inflammatory process associated with juxtaarticular deposits of calcium hydroxyapatite, and has been described in the vicinity of almost every joint in the hand and wrist.[37] Patients may present with significant pain and erythema so severe that it simulates infection, which is not usually seen with trigger finger. Locking or triggering of the finger is also not observed with calcific peritendinitis or periarthritis. Radiographs may demonstrate fluffy calcification in the area of inflammation, but this is not usually apparent until a few days after the onset of symptoms.

Tenosynovitis (non-infectious):—

Pain, tenderness, and swelling along the flexor tendon in the hand may be associated with an underlying inflammatory arthritis such as rheumatoid arthritis or reactive arthritis. It can be differentiated from trigger finger in that tenosynovitis often leads to swelling and pain along the long axis of the affected tendon, joints, or finger, resulting in limited or painful extension of the involved digit. Tenosynovitis improves with treatment of the underlying arthritis with medications such as nonsteroidal antiinflammatory drugs (NSAIDs), disease-modifying antirheumatic drugs (DMARDs), and systemic glucocorticoids.

PROGNOSIS

The prognosis in trigger finger is very good; most patients respond to corticosteroid injection with or without associated splinting. Some cases of TF may resolve spontaneously and then reoccur without obvious correlation with treatment or exacerbating factors.

Freiberg et al found a greater success rate for TF injection therapy when the treatment was used in patients in whom an examiner could palpate a discrete, rather than a diffuse, nodular consistency in the flexor sheath.[38] Digits with a discrete, palpable nodule had a 93% success rate with a single injection of triamcinolone at 3 months' follow-up, whereas digits with a diffuse pattern had a 52% failure rate.

Marks and Gunther reported an 84% success rate in trigger digits and a 92% success rate in trigger thumbs following a single injection of triamcinolone.[39]

Using sonoelastography, a newer technique for quantitative assessment of the stiffness of soft tissues, one group noted that the causes for snapping in TF were increased stiffness and thickening of the A1 pulley. Three weeks after corticosteroid injection, the pulley thickness and the ratio of subcutaneous fat to the pulley both decreased; snapping disappeared in all patients studied.

Griggs et al reported an overall success rate of 50% for steroid injection in patients with DM.[40] Patients with insulin-dependent diabetes had a higher incidence of multiple digit involvement and required surgical release more frequently than did patients who were not insulin-dependent.[41-42]

MANAGEMENT:

Approach Considerations

Early series recommended surgical treatment of trigger finger (TF) as straightforward and highly effective, while regarding prolonged conservative treatment as unreliable and expensive. Subsequent series documented poor results from surgical treatment in 7-9% of cases.

In 1972, Lapidus reversed his previous recommendation for operative treatment of TF after he and Guidotti reported uniformly good results following a single injection of prednisolone into the tendon sheath.[43] Rhoades et al subsequently reported a 72% success rate in a series of 53 digits following injection and immobilization.[44]

Injection therapy is now generally agreed to be the first line of management. Surgery is reserved for individuals in whom injection treatment has failed or in whom other pathology, particularly rheumatoid arthritis (RA), is suspected to be causing triggering that cannot be treated conservatively.[45] No absolute contraindications exist for surgical management.

In May 2014, the European HANDGUIDE Group published a guideline for multidisciplinary treatment of trigger finger.[46] By consensus, suitable treatment options were considered to include the following:

- Orthoses (splinting)

- Corticosteroid injections
- Corticosteroid injections plus use of orthoses

Surgery

Severity and duration of disease and prior treatments received were judged to be the primary factors influencing choice of therapy.[46]

Conservative treatment

Most trigger digits in adults can be managed successfully with local steroid injections and splinting.[47] Oral or topical pharmacologic measures have not been demonstrated to be effective.

The outcome of conservative treatment for pediatric trigger thumb is somewhat controversial.[48] A report by Baek et al on the natural history of this condition demonstrated that after a follow-up period of 5 years or more in patients who received no treatment for pediatric trigger thumb, complete resolution of flexion deformity occurred in 66 out of 87 thumbs (75.9%), and partial improvement occurred in the remaining 21 thumbs.[49]

Another study, by Lee et al, reported that extension splinting for 12 weeks led to improvement in 71% of thumbs, compared with 23% improvement in patients not receiving any treatment.[50]

Surgical release

The chief indications for surgical management of TF are as follows:

- Failure of splinting and/or injection treatment
- Irreducibly locked TF
- Trigger thumb in infants - Without surgical release, these infants are likely to develop a fixed flexion deformity of the interphalangeal (IP) joint

Although the results of percutaneous release are well established, the open technique is absolutely essential for the thumb or little finger or in the presence of proximal interphalangeal (PIP) contractures. Percutaneous release should be reserved for the index, middle, and ring fingers.[51]

In a study from Oxford comparing percutaneous and open surgical methods, the two approaches displayed similar effectiveness, and both proved superior to conservative corticosteroid-injection treatment with regard to trigger cure and relapse rates.[52]

In children, triggering has varying causes. Release of the A1 pulley alone does not always correct the problem. Additional treatment (eg, resection of one or both limbs of the flexor digitorum superficialis [FDS] tendon, A3 pulley release) may be required and is recommended in RA tenosynovitis.[53-54]

In infants, the nodule on the flexor pollicis longus (FPL) tendon can be resected with good results. Corticosteroid injections are generally not helpful in these cases of trigger thumb.

Pregnant patients

Splinting and local corticosteroid injection can be performed if the patient is pregnant. Surgical release of the A1 pulley is generally an elective procedure and is usually deferred until after delivery.

Elderly patients

In elderly patients with a history of gastrointestinal problems or other complications from nonsteroidal anti-inflammatory drugs (NSAIDs), consider cyclo-oxygenase-2 (COX-2) inhibitors if oral NSAIDs are needed.

Consultations

Surgical consultation for operative treatment may be required. Typically, such procedures are performed by an orthopedic hand surgeon or a plastic surgeon.

CORTICOSTEROID INJECTION INTO TENDON SHEATH

Corticosteroid injection in the area of tendon sheath thickening is considered to be the first-line treatment of choice for TF.[55-56] Research in 2009 concluded that the most successful and cost-effective management strategy for TF is the algorithm of two steroid injections prior to surgical intervention, if needed.[57]

A variety of preparations have been used—most commonly prednisolone, dexamethasone, and triamcinolone—in the steroid injection treatment of TF, and most are uniformly successful in relieving symptoms.

A highly satisfactory rate of success can be predicted in female patients and in patients with single digit involvement, short duration of symptoms (ie, < 4 months), no associated conditions (eg, RA, diabetes mellitus [DM]), or a discrete, palpable nodule. (Patients with RA or DM seem to be more resistant to injection treatment.)[58]

Procedure

The author's technique for steroid injection is as follows. A mixture of triamcinolone, 1% lidocaine, and 0.5% bupivacaine is used, in a ratio of 2:1:1, respectively; adrenaline is not used. The nodule in the palm is well localized and circled out using an indelible skin marker. The procedure is performed in an office setting, using strict aseptic precautions, with alcoholic povidone-iodine used for injection-site preparation. Ethyl chloride is used only if requested; frequently, it is unnecessary, and most patients tolerate this procedure quite well

A 26-gauge needle is introduced in a proximal-to-distal direction in the nodule, making an angle of 45° with the palm (Fig. 5). The needle enters the nodule with a distinct grating sensation; positioning of the needle is verified by asking the patient to move the digit when it is possible to clearly observe the needle moving with the digit (Fig. 6)

The syringe with the steroid preparation then is attached to the needle. Attempting to inject the drug with light pressure confirms the intratendinous location of the needle. Do not inject the solution if significant resistance to injection flow is noted, because this may indicate that the needle tip is in the tendon rather than just within the tendon sheath. The needle is withdrawn very carefully until a give-way sensation is felt, indicating that the tip of the needle is out of the tendon and in the sheath. The preparation is then injected.

A small water-impermeable dressing is applied. The patient is actively encouraged to move the digit; in most cases, the triggering is relieved

(Carlson and Curtis prefer a midaxial injection at the level of the midproximal phalanx as a simple and painless way to access the flexor sheath for the purpose of injection.)[59]

A follow-up appointment is made for 3-4 weeks after the treatment. Splinting is not used routinely for these cases, though a hand-based MP-block Orthoplast splint has been described as useful.

Although injection treatment has long been administered by "feel" and experience, research suggests that using ultrasonographically guided steroid injection may maximize the injection's accuracy and, consequently, its beneficial effects in the treatment of trigger digits.[60-61]

No major complications from injection treatment are noted. A transient rise in blood and urine sugar levels is common in patients with diabetes. Advise these patients that this is likely to occur. Theoretically, repeated steroid injections could cause attrition and/or rupture of tendons, but only 1 such case has been reported to date.[62]

Fig. 5

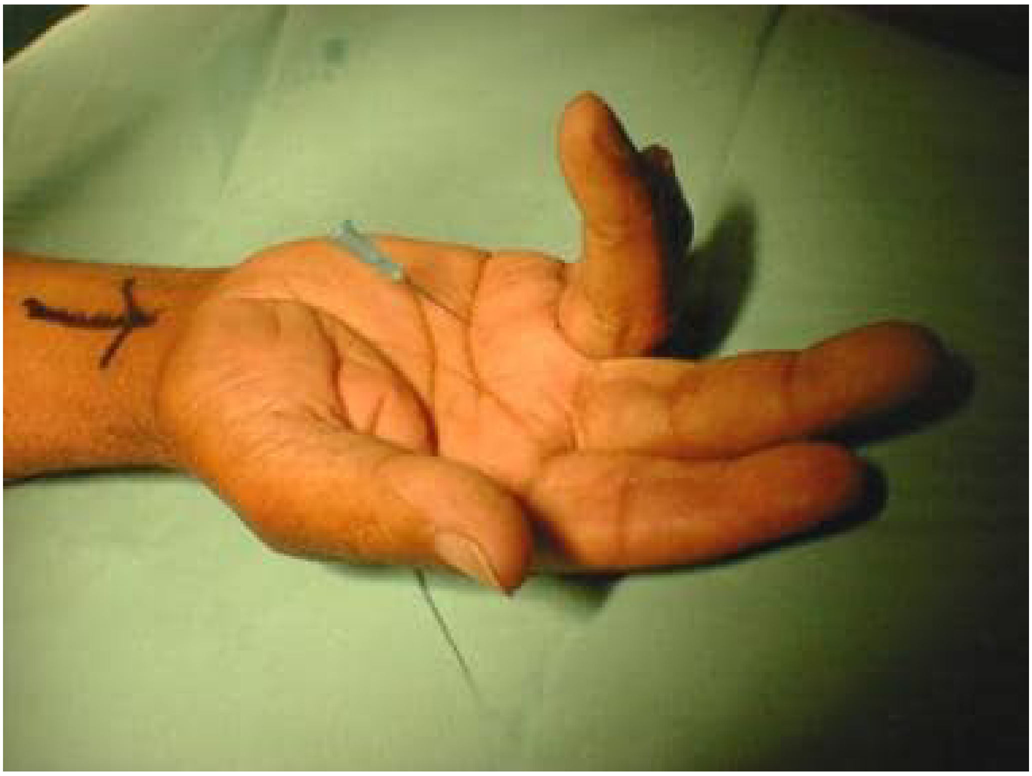

Fig. 6

Pain relief

While corticosteroid injections into the palm are considered highly effective in treating TF, the injection itself may be significantly painful. A study that compared conventional injection with injection preceded by median and ulnar nerve blocks performed at the wrist found that of the study’s 19 patients, who were treated with 47 total injections, 88% preferred to have the median and ulnar nerve block prior to the injection.[63]

Both study groups, however, had excellent resolution of TF, with excellent resolution being defined in the study as an asymptomatic hand (without triggering) and a pain score on the 0-10 cm Visual Analog Scale of less than 2 cm.[63]

Repeat injections

A second corticosteroid injection may be performed 3-4 weeks after the first one. If two or perhaps three injections fail to provide adequate resolution, consider referring the patient for surgical release. Repetitive injections theoretically increase the likelihood of tendon rupture, although such a risk was not found in Anderson's study of repeated injections for TF.[64]

Alternative injection techniques

Proximal phalanx technique

Another injection method, the proximal phalanx technique, allows for injection directly into the tendon sheath through the palmar surface of the midproximal phalanx. Injections performed this way were found to be less painful than injecting the flexor tendon sheath directly over the metacarpal head. There was no statistically significant difference in the rate of recurrent pain between the two injection methods.[65]

Subcutaneous injection

Although corticosteroid injection has traditionally been administered into the tendon sheath (but not into the tendon itself), studies seem to indicate that subcutaneous injection may be as effective as the intrasheath approach.[66-68] Additionally, in some cases, steroid injection into the subcutaneous tissue seems to result in better clinical outcomes than does injection into the sheath alone.[66]

Splinting

Custom-made splinting of the metacarpophalangeal (MCP) joint is another conservative treatment, used in patients who do not wish to undergo a steroid injection or as an adjuvant to injection. Typically, a custom-made splint is used to hold the MCP joint of the involved finger at 10-15° of flexion, leaving the PIP and distal interphalangeal (DIP) joints free. The average length of splinting is 6 weeks. In patients with symptoms longer than 6 months, splinting as a sole treatment strategy does not seem to eliminate the triggering events.

Although traditionally splinting has not been thought to be an effective treatment for TF, one study of thermoplastic splinting of MCP joint flexion showed improvement in stenosing tenosynovitis, the numeric pain rating scale, and the number of triggering events and also demonstrated an overall perceived participant improvement in symptoms.[69] Another study determined that 87% of patients who wore custom-made, thermoplastic orthoses for 8-10 weeks did not require an injection or surgical intervention in the 1-year follow-up after institution of the orthoses.[70]

SURGICAL RELEASE

Trigger digits that fail to respond to two or perhaps three injections may require surgical treatment, including dissection of the nodule on the tendon and surgical release of the A1 pulley, under local anesthesia.

The benefits of operative treatment of trigger finger and trigger thumb were outlined in three studies of surgical pulley release.

Between 1994 and 2004, Li et al treated seven children (nine thumbs; three right, two left, two bilateral) for trigger thumb with hyperextensible MCP anomaly (>60°) by surgical release of the first anular pulley (A1 pulley) and proximal advancement of the MCP volar plate. The patients (four girls, three 3 boys), who had a mean age of 46 months at surgery (range, 26-82 months), were observed over a mean follow-up period of 64 months (range, 1-8 years).

All patients in the study at last follow-up had returned to full activity without limitation or pain, and none of the patients had a recurrence of triggering or MCP hyperextension deformity, demonstrating, according to the authors, that trigger thumb with concomitant MCP hyperextension deformity can be treated in children by A1 pulley release and advancement of the volar plate.[71]

In a study of 93 trigger thumbs in 83 patients, Chao et al compared the results of miniscalpel-needle percutaneous release with those of steroid injection. At 12 months, 44 of the 46 trigger thumbs treated with the miniscalpel-needle release had satisfactory results (measured by visual analogue pain scale and patient satisfaction), but only 12 of 47 thumbs treated with steroid injection had satisfactory results. No nerve injuries occurred in either group.[72]

Trigger thumb in children almost always calls for surgical management. Trigger thumb in an adult not responding to corticosteroid tendon sheath injection needs surgery. The technique of release itself is irrelevant. Open and not percutaneous surgery is the norm for trigger thumb in children and adults alike, since the neurovascular bundles in the thumb are closer to the midline than in other digits. A single series (quoted above) comparing the efficacy of percutaneous surgery vis-a-vis a corticosteroid injection found surgery still to be more effective than injection treatment, but this technique of surgical release itself is not de rigueur.

Lange-Rieb et al presented long-term results of open operative treatment of TF and trigger thumb in adults. Of the operations performed, 210 (76%) were for a single-digit release and 76 (24%) for multiple digits. All operations were performed under tourniquet control with local anesthesia as outpatient procedures using a transverse incision just distal to the distal palmar crease or on the flexor crease of the thumb at the MCP joint. At latest follow-up (average, 14.3 years), 234 patients were evaluated, with no complaints, and there were no serious complications, such as nerve transection or bowstringing, or recurrence.[73]

Preparation for surgery

Preoperative considerations include the following:

- o Only digits that actively trigger must be considered for operative release

- Neither PIP contracture nor thumb triggering is suitable for percutaneous release, and the A1 pulley always is transected under direct vision

- Patients with PIP joint contractures undergo a period of hand therapy and splinting prior to the procedure

- A tourniquet always is used to obtain a clean operative field
- Approximately 4-5 mL of 1% lidocaine is used to infiltrate the skin overlying the A1 pulley, with injection performed deeper to the tendon sheath
- The transverse incision is marked with a skin marker corresponding to the digit to be surgically treated
- The proximal edge of the A1 pulley coincides almost exactly with the distal palmar crease in the fourth and fifth rays, with the proximal palmar crease in the index and with the halfway point between the two creases in the middle finger

Operative details

The MCP joint is hyperextended to displace the neurovascular structures dorsally, minimizing the risk of injury.

A transverse incision measuring 1-1.5 cm is made over the involved metacarpal head. Blunt dissection is used to spread the subcutaneous fat and expose the tendon sheath.

The proximal edge of the A1 pulley is identified, and a scalpel blade is used to divide the entire A1 pulley in the midline under vision. Care is taken to avoid incising too distally and risk cutting into the A2 pulley, which can result in bowstringing. A study suggests that the proximal part of the A2 pulley can be safely incised if the release of the A1 pulley in isolation does not result in relief of triggering.[74] This is still experimental and is best left to hand or plastic and reconstructive surgeons.

The patient is asked to actively move the digit to confirm full release. Meticulous hemostasis is achieved with a bipolar cautery, and the wound is closed with two or three skin sutures. The hand is left free, and motion is encouraged immediately following the procedure.

If a percutaneous approach is favored, a pair of blunt-tipped, fine scissors is introduced through the incision, and the A1 pulley is transected. Care is taken not to drift too distally. Disappearance of a grating sensation indicates complete section of the pulley through a separate, distal oblique incision.

A study by Rogo-Manaute et al showed that it is possible to use ultrasonographically guided percutaneous release to achieve a success rate of 100%. With adequate anatomic knowledge, technical training, and a basic ultrasound machine, sonographically directed A1 pulley release can be performed safely and successfully, thus offering an alternative to conventional open technique.[75]

On rare occasions, sectioning the A1 pulley does not relieve triggering, indicating that the A3 pulley might be involved. If that is the case, the A3 pulley requires division. This percutaneous technique as described here usually applies to most cases of triggering, exceptions being surgery for trigger thumb in children and triggering involved in conditions like RA, in which the nodule formation may be distal to the A1 pulley and for which open surgery may be required.

THE ORIGINAL STUDY

OBJECTIVE

- To compare the frequency of cure and relapse of percutaneous A1 pulley release with conventional open surgery in treatment of the trigger finger

OPERATIONAL DEFINITIONS

Trigger Finger:

- Trigger finger is a condition that causes pain, stiffness, and a sensation of locking or catching when you bend the straighten your finger. The condition is also known as "stenosing tetenosynovitis". The finger and thumb are most often affected by trigger finger, but it can occur in the other fingers, as well. When the 1 thumb is involved, the condition is caller "trigger thumb". This is a clinical diagnosis and doctor diagnosed trigger by tenderness over flexor tendon sheath in the palm in the hand.

Cure of the trigger:

- It was considered as the remission of the trigger, cessation of blockage (bending) of the finger, and the free flow of its movement and it was considered on 1^{st} post procedure day measured on clinical Examination.

Relapse of the trigger:

- It was classified as the relapse or return of blockage of the finger within the 03 months of post-study follow-up.

HYPOTHESIS

- There is a difference on frequency of cure on relapse in treatment of trigger finger by percutaneous A1 pulley with conventional open surgery

MATERIAL & METHODS:

STUDY DESIGN:

- Randomized Controlled Trial

SETTINGS:

- Orthopaedics Department, Benazir Bhutto Hospital, Rawalpindi

DURATION OF STUDY:

- 6 months after the approval of synopsis

  From: _______________ to _______________

SAMPLING TECHNIQUE:

- Simple random sampling

SAMPLE SIZE:

- Sample size of 60 (30 in each group) was calculated by using 95% confidence level with 80% power of test and by taking expected cure rate in percutaneous release is 93.75% and expected cure rate in open surgery group is 100%[2]

SAMPLE COLLECTION:

Inclusion criteria:

- Patients with diagnosed trigger finger by the consultant by clinical examination (grade II, III, IV annexure C)
- Both genders included
- Age: 18-60 years of age

Exclusion criteria:

- Who were treated with a previous operation or a corticosteroid injection for their disease and those who were suffering by inflammatory arthritis, tumor or autoimmune disease
- Multiple trigger fingers trying to evaluate our study groups strictly using the model one patient-one trigger finger
- Insulin-dependent diabetes mellitus and a history of other tendon related pathology of the upper extremity, although it is known that they might be linked with a higher rate of treatment failure

DATA COLLECTION PROCEDURE:

After taking written informed consent, patients meeting the inclusion criteria were randomly assigned to either group; Group A (Percutaneous release) and Group B (Open surgery) by lottery method. Group A was treated with a percutaneous release of the affected first annular pulley under local anesthesia and without any corticosteroid injection. In addition, group B underwent a conventional open surgical release of the A1 pulley, through a 10-15 mm incision. Demographic information (including name, age and gender) was also recorded.

Technique:

The technique in Group A was a guided local anesthetic injection, proximally to the metacarpal-phalangeal joint. The infiltration was done under sterile conditions (sterilization of the skin, coverage of the ultrasound probe with sterile pad, use of appropriate gel) and then the physician was inserted percutaneously---through a negligible section <1mm—an ophthalmic corneal/sclera V-Lance knife (Alcon, Novartis company), over flexor tendons (Verdan's zone 3, proximally to the A1 pulley) and towards their longitudinal axis.

Then, the knife was advanced distally, just below A1 pulley and passed palmary so as to loosen the thicken pulley (the interesting part). Thus, after having withdrawn the V-Lance knife, a thin hook with a long neck was introduced under the — now extended - Al 'volley. The hook was penetrated the annular ligamentous structure facing palmary in order to protect the flexor tendons and subsequently removed proximally (in a steady quick move) carrying along and dissecting the Al pulley.

Intra-operatively and right after the performed dissection, each patient was clinically evaluated for the achieved resolution of the triggering. All patients were estimated with the completion of Q-DASH score before and after the operation at 4 weeks. Resolution of triggering (primary variable of interest) was expressed as the "success rate" per digit Detail Perforrna is attached (Annexure A). Cure at 1st post op day and after 3 months, relapse was ---within and after 3 months.

DATA ANALYSIS:

Data was analyzed using SPSS version 22. Quantitative variable i.e. age and Q-Dash Score pre and post operatively presented with mean and Standard deviation. Qualitative variables like gender, age categories, type of trigger finger and Post-treatment cure and relapse was presented with frequency and percentage.

Effect modifiers like age, gender and type of trigger finger according to green classification was controlled by stratification. Post Stratification Chi-tat was applied and also Chi square test was applied to compare cure and relapse between 2 groups p-value 0.05 was significant. Results were presented in tabulated or graphical forms.

RESULTS

A total of 60 cases (30 in each group) fulfilling the selection criteria were enrolled to compare the frequency of cure and relapse of percutaneous A1 pulley release with conventional open surgery in treatment of the trigger finger.

Age distribution shows that 80%(n=26) in Group-A and 86.67%(n=24) in Group-B were between 18-40 years of age whereas 20%(n=4) in Group-A and 13.33%(n=6) in Group-B were between 41-60 years of age, mean$\pm$sd was calculated as 33.7$\pm$6.05 in Group-A and 34.4$\pm$5.85 years in Group-B. (Table No.1)

Gender distribution shows that 70%(n=21) in Group-A and 53.33%(n=16) in Group-B were male whereas 30%(n=9) in Group-A and 46.67%(n=14)in Group-B were females. (Table No. 2)

Type of finger trigger was recorded as 33.33%(n=10) in Group-A and 36.67%(n=11) in Group-B had II, 36.67%(n=11) in Group-A and 40%(n=12) in Group-B had III while 30%(n=9) in Group-A and 23.33%(n=7) in Group-B had IV type of finger trigger. (Table No. 3)

The frequency of cure and relapse of percutaneous a1 pulley release with conventional open surgery in treatment of the trigger finger was recorded as 86.67%(n=26) in Group-A and 96.67%(n=29) in Group-B cure while 13.33%(n=4) in Group-A and 3.33%(n=1) in Group-B were relapse, p value was 0.16. (Table No. 4)

Effect modifiers like age, gender and type of trigger finger according to green classification was controlled by stratification. (Table No. 5-7)

TABLE No. 1
AGE DISTRIBUTION
(n=60)

Age (in years)	Group-A (n=30)		Group-B (n=30)	
	No. of patients	%	No. of patients	%
18-40	26	80	24	86.67
41-60	4	20	6	13.33
Total	30	100	30	100
Mean±SD	33.7±6.05		34.4±5.85	

TABLE No. 2
GENDER DISTRIBUTION
(n=60)

Gender	Group-A (n=30)		Group-B (n=30)	
	No. of patients	%	No. of patients	%
Male	21	70	16	53.33
Female	9	30	14	46.67
Total	30	100	30	100

TABLE No. 3
TYPE OF FINGER TRIGGER
(n=60)

Type of finger trigger	Group-A (n=30)		Group-B (n=30)	
	No. of patients	%	No. of patients	%
II	10	33.33	11	36.67
III	11	36.67	12	40
IV	9	30	7	23.33
Total	30	100	30	100

TABLE No. 4
FREQUENCY OF CURE AND RELAPSE OF PERCUTANEOUS A1 PULLEY RELEASE WITH CONVENTIONAL OPEN SURGERY IN TREATMENT OF THE TRIGGER FINGER
(n=60)

Cure and relapse	Group-A (n=30)		Group-B (n=30)	
	No. of patients	%	No. of patients	%
Cure	26	86.67	29	96.67
Relapse	4	13.33	1	3.33
Total	30	100	30	100

P value=0.16

TABLE No. 5
STRATIFICATION FOR CURE AND RELAPSE
WITH REGARDS TO AGE
(n=60)

Age (in years)	Group	Cure/Relapse		P value
		Cure	Relapse	
18-40	A(n=26)	22(84.62%)	4(15.38%)	0.351
	B(n=24)	23(95.83%)	1(4.17%)	
41-60	A(n=4)	6(100%)	0	--
	B(n=6)	4(100%)	0	

TABLE No. 6
STRATIFICATION FOR CURE AND RELAPSE
WITH REGARDS TO GENDER
(n=60)

Gender	Group	Cure/Relapse		P value
		Cure	Relapse	
Male	A(n=21)	17(80.95%)	4(19.05%)	0.118
	B(n=16)	16(100%)	0	
Female	A(n=9)	9(100%)	0	1.00
	B(n=14)	14(100%)	0	

TABLE No. 7
STRATIFICATION FOR CURE AND RELAPSE
WITH REGARDS TO TYPE OF FINGER TRIGGER
(n=60)

Category	Cure/Relapse	Group		P value
		Group-A	Group-B	
II	Cure	10(100%)	11(100%)	--
	Relapse	0	0	
III	Cure	9(81.82%)	12(100%)	0.217
	Relapse	2(18.18%)	0	
IV	Cure	7(77.78)	6(85.71%)	1.00
	Relapse	2(22.22%)	1(14.29%)	

DISCUSSION

Stenosing tenosynovitis or trigger finger is characterised by pain, swelling, movement limitation, and a triggering sensation of the finger. The primary pathology is thickening of the A1 pulley, with resultant entrapment of the flexor tendon. Percutaneous release of the A1 pulley results in a low complication rate and high patient satisfaction.

There is lack of randomized clinical trials in Pakistan, comparing open surgery to percutaneous release, however, in the present randomized controlled trial; we compared the frequency of cure of percutaneous A1 pulley release with conventional open surgery in terms of ability to correct the trigger finger so that better management be planned.

In our study, mean age was calculated as 33.7±6.05 in Group-A and 34.4±5.85 years in Group-B, 70%(n=21) in Group-A and 53.33%(n=16) in Group-B were male whereas 30%(n=9) in Group-A and 46.67%(n=24)in Group-B were females. The frequency of cure and relapse of percutaneous a1 pulley release with conventional open surgery in treatment of the trigger finger was recorded as 86.67%(n=26) in Group-A and 96.67%(n=29) in Group-B cure while 13.33%(n=4) in Group-A and 3.33%(n=1) in Group-B were relapse, p value was 0.16.

Nikolaou et al[2] reported 93.75% cure and 6.25% relapse with USG percutaneous release and 100% cure and 0% relapse with open surgery, cosmetic results found excellent or good in 87.5% in percutaneous release patients, while only 56.25% patients were evaluated as faire or poor in open surgery. The relapse and cure rate in our study is in agreement with Nikolaou et al.

Chin-JungLin and others[76] compared the short-term (3 months) and long-term (2 years) outcomes between open surgical release and percutaneous needle release of trigger finger. It was concluded that the percutaneous release method to release trigger finger does not have a better long-term satisfaction rate than the open release approach, although percutaneous release has a significantly better short-term satisfaction rate. However, we recorded outcome on short term basis and found no significant difference with conventional method.

Marij Zahid and others[77] compared the outpatient Percutaneous Release of Trigger Finger and concluded that percutaneous release of trigger finger with needle was not only associated with excellent functional outcome and recovery in terms of patient satisfaction and range of finger motion three months post-procedure but also was found to be cost effective.

The percutaneous surgical release technique performed by Eastwood et al[78] is a convenient, minimally invasive, economical method with a very low complication rate, and is becoming more popular than open surgery.

Mohsen[79] in his study, reported 97% success rate of percutaneous release in 40 trigger digits, the thumb being the most common digit, similar to our study which showed 100% successful release and the thumb was also the most common digit involved. Sahu et al[80] reported successful results in 95.6% patients (excellent in 82.6% and good in 13%).

We found percutaneous technique for release of trigger finger is safe, cost effective technique with significant patient satisfaction. It may performed in the clinic, just requires an anesthetic and a disposable 18 gauge needle and shows promising results while on other hand open release requires a day care procedure, use of sterilized equipment, skin incision and a suture. With a resource constraint country, percutaneous release of trigger finger proves to be a highly cost-effective method. The only pitfall of percutaneous technique is its blind nature but with very few complications. This study impels the reviewers and opens the grounds for further elaborated and extensive studies in future.

CONCLUSION

- We found no significant difference in frequency of cure and relapse of percutaneous A1 pulley release with conventional open surgery in treatment of the trigger finger.

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